MILITARY SPECIFICATION

UNIVERSAL MICROWAVE CALIBRATORS (UMC): MX–3297/USM–129, MX–3169/USM–129 THROUGH MX–3175/USM–129, MX–3297A/USM–129, MX–3169A/USM–129 THROUGH MX–3175A/USM–129

1. SCOPE

1.1 Scope. This specification covers a series of Universal Microwave Calibrators (UMC's) and associated plug-in signal sources in the frequency range of 0.3 kilomegacycles per second (kmcps) to 40.0 kmcps. These Universal Microwave Calibrators (UMC's) form part of Microwave Calibration Test Sets AN/USM–129, AN/USM–129A and AN/USM–129B.

1.2 Classification. The UMC's covering the frequency range of 0.3 to 40.0 kmcps shall be designated by the nomenclature given in table I. Plug-in signal sources are required for the UMC's in the frequency range of 18.0 to 40.0 kmcps. The plug-in signal sources shall be as designated in table II.

TABLE I. *Universal microwave calibrators (UMC's)*

Frequency range (kmcps)	For AN/USM–129		For AN/USM–129A and 129B	
	Designator	*Ordnance Part No.*	*Designator*	*Ordnance Part No.*
0.3 — 1.0	MX–3297/USM–129	10519217	MX–3297A/USM–129	10528436
0.95— 2.0	MX–3169/USM–129	10518923	MX–3169A/USM–129	10528437
1.95— 4.0	MX–3170/USM–129	10518925	MX–3170A/USM–129	10528438
3.95— 8.2	MX–3171/USM–129	10518924	MX–3171A/USM–129	10528439
8.2 —12.4	MX–3172/USM–129	10519363	MX–3172A/USM–129	10528440
12.4 —18.0	MX–3173/USM–129	10519495	MX–3173A/USM–129	10528441
18.0 —26.5	MX3174/USM–129	10519306	MX–3174A/USM–129	10528442
26.5 —40.0	MX3175/USM–129	10519246	MX–3175A/USM–129	10528443

TABLE II. *Plug-in signal sources for UMC's*

Frequency range (kmcps)	For AN/USM–129		For AN/USM–129A and 129B	
	Designator	*Ordnance Part No.*	*Designator*	*Ordnance Part No.*
18.0—22.0	MX3176/USM–129	10519319	MX4254/USM–129A	10528444–1
22.0—25.0	MX3177/USM–129	10519302	MX4255/USM–129A	10528444–2
24.0—27.8	MX3178/USM–129	10519247	MX–4483/USM–129A	10528446–1
			MX–4256/USM–129A	10528445
27.8—32.2	MX3179/USM–129	10519248	MX4257/USM–129A	10528446–2
32.0—37.5	MX3180/USM–129	10519249	MX4258/USM–129A	10528446–3
35.0—40.0	MX3181/USM–129	10519250	MX4259/USM–129A	10528446–4

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–I–45208 —Inspection Requirements; General specification for

MIL–F–46349–(ORD) —Frequency Meters, Coaxial

MIL–C–46350–(ORD) —Combination Case, Multipurpose, for Test Set, Microwave Calibration

DRAWINGS

ORDNANCE CORPS

For AN/USM–129

D10518923 —Calibrator, Microwave, Universial 0.95–2.0 kmc

D10518924 —Calibrator, Microwave, Universal, 3.95–8.2 kmc

D10518925 —Calibrator, Microwave, Universal, 1.95–4.0 kmc

D10519217 —Calibrator, Microwave, Universal, 0.3–1.0 kmc

D10519246 —Calibrator, Microwave, Universal, 26.5–40.0 kmc

D10519247 —Plug-in R–F Head, A-band

D10519248 —Plug-in R–F Heads, A-band

D10519302 —Plug-in R–F Heads, K-band

D10519306 —Calibrator, Microwave, Universal, 18.0–26.5 kmc

D10519363 —Calibrator, Microwave, Universal, 8.2–12.4 kmc

For AN/USM–129A and 129B

D10519495 —Calibrator, Microwave, Universal, 12.4–18.0 kmc

For AN/USM–129A and 129B

E10528436 —Calibrator, Microwave, Universal, 0.3–1.0 kmc

E10528437 —Calibrator, Microwave, Universal, 0.95–2.0 kmc

E10528438 —Calibrator, Microwave, Universal, 1.95–4.0 kmc

E10528439 —Calibrator, Microwave, Universal, 3.95–8.2 kmc

E10528440 —Calibrator, Microwave, Universal, 8.2–12.4 kmc

E10528441 —Calibrator, Microwave, Universal, 12-4–18.0 kmc

E10528442 —Calibrator, Microwave, Universal, 18.0–26.5 kmc

E10528443 —Calibrator, Microwave, Universal, 26.5–40.0 kmc

F10528444–1 —Plug-in R–F Head, K-band

F10528444–2 —Plug-in R–F Head, K-band

D10528445 —Plug-in R–F Head, K-band and A-band

D10528446–1 —Plug-in R–F Head, A-band

D10528446–2 —Plug-in R–F Head, A-band

D10528446–3 —Plug-in R–F Head, A-band

D10528446–4 —Plug-in R–F Head, A-band

For AN/USM–129, 129A and 129B

A10519341 —Calorimeter

A10519351 —Calorimeter

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A10519472 —Calibrator, Attenuation Assembly
A10528591 —Power Meter Amplifier, Calorimetric

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Initial Production Sample. Unless otherwise directed by the contracting officer, an initial production sample shall be required. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production.

3.2 Materials. Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction. The UMC's shall be manufactured in accordance with the applicable drawings in 2.1, as required, and all drawings referenced thereon. The coaxial frequency meters and combination cases which form a part of the UMC's shall comply with the appropriate specications in 2.1. The requirements of the specification are detailed only to the extent considered necessary to obtain the desired electrical and mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities.

3.4 Electrical

3.4.1 *Frequency range.* The frequency range of the UMC's and of the plug-in signal sources shall be as specified in tables I and II.

3.4.2 *Output power.* The output power of the UMC's shall be as specified in Table III.

3.4.3 *Voltage Standing Wave Ratio (VSWR).* The maximum permissible VSWR at the ports indicated shall be as specified in table III.

TABLE III. *Power output and VSWR of UMC's*

Designator	Min. RF power output at SIG. SOURCE OUTPUT port (milliwatts)	Maximum VSWR at		
		SIG. SOURCE OUTPUT port	FILTER CAL INPUT port	FREQ. CAL INPUT port
MX–3297/USM–129	10.0	3.0	.....	3.0
MX–3297A/USM–129				
MX–3169/USM–129	5.0	1.8	4.0	1.8
MX–3169A/USM–129				
MX–3170/USM–129	5.0	2.1	4.0	2.1
MX–3170A/USM–129				
MX–3171/USM–129	5.0	2.1	4.0	2.1
MX–3171A/USM–129				
MX–3172/USM–129	5.0	1.7	3.0	1.5
MX–3172A/USM–129				
MX–3173/USM–129	5.0	1.7	3.0	1.5
MX–3173A/USM–129				
MX–3174/USM–129	5.0	1.7	3.0	1.5
MX–3174A/USM–129				
MX–3175/USM–129	5.0	1.7	3.0	1.5
MX–3175A/USM–129				

3.4.4 *Crystal Current.* The crystal current shall be 3 microamperes minimum with 3 milliwatts of RF power available at SIG. SOURCE OUTPUT port.

3.4.5 *Rise and Fall Time of RF Notch.* The rise and fall time of the RF notch shall be 2 microseconds maximum for MX–3297/USM–129 and MX–3297A/USM–129 and 1.5

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microseconds for all other UMC's, for a 5 microseconds wide notch at 1,000 notches per second when modulating with a pulse of .015 microseconds maximum rise and fall time.

3.4.6 *Detected Notch Amplitude on Oscilloscope.* The detected notch amplitude on the oscilloscope shall be one centimeter minimum with 0.5 milliwatt of RF power available at the SIG. SOURCE OUTPUT port for MX–3175/USM–129 and MX–3175A/USM–129 and 1.0 milliwatt for all other UMC's when the oscilloscope sensitivity is set at 0.01 volts/cm.

3.4.7 *Calibration Curve.* A calibration curve of attenuation (in db) versus frequency between the RF PULSE INPUT and the SIG. SOURCE OUTPUT ports shall be provided.

3.5 Environmental.

3.5.1 *Operating.*

3.5.1.1 *Temperature.* Prior to, during, and after being exposed and thermally stabilized at ambient temperatures of plus 32° and plus 110° Fahrenheit, the UMC shall meet the requirements of 3.4.

3.5.1.2 *Humidity.* Prior to, during and after being subjected to the procedure specified in 4.7.1.2, the UMC shall meet the requirements of 3.4.

3.5.1.3 *Altitude.* The UMC shall be capable of operating at a reduced pressure of 20.58 inches of mercury (equal to an altitude of 10,000 feet) for 4 hours.

3.5.2 *Nonoperating.*

3.5.2.1 *Temperature.* Prior to and after being subjected to the procedure specified in 4.7.2.1, the UMC shall meet the requirements of 3.4.

3.5.2.2 *Altitude.* Prior to and after being subjected to the procedure specified in 4.7.2.2, the UMC shall meet the requirements of 3.4.

3.5.2.3 *Vibration.* After being subjected to the procedure specified in 4.7.2.3, the UMC shall show no physical damage and shall meet the requirements of 3.4.

3.5.2.4 *Shock.* After being subjected to the procedure specified in 4.7.2.4, the UMC shall show no physical damage and shall meet the requirements of 3.4.

3.6 Workmanship. The UMC's and their associated plug-in signal sources, including all parts and accessories, shall be manufactured and finished in a thoroughly workmanlike manner and in accordance with good commercial practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilities. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 General Requirements. Inspection requirements and procedures shall be in accordance with Specification MIL–I–45208.

4.3 Initial Production Sample. Prior to production, the contractor shall submit the initial production sample in accordance with 3.1 for inspection by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.3.1 *Production Sample Failure.* Should the production sample fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. Where corrections in the process have been

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made, a new production sample shall be examined and tested by the contractor under the surveillance of the government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

4.4 Material. Control of material shall be in accordance with the test data requirements of Specification MIL–I–45208.

4.5 Inspection of Components. The contractor shall ascertain that prior to assembly all components procured under separate specifications or drawings, have been inspected, tested and accepted in accordance with their respective specifications or drawings.

4.6 Electrical.

4.6.1 *Frequency Range.* When the UMC is oscillated at each of the test frequencies in 4.6.8, an indication shall be obtained on the crystal current meter of the UMC.

4.6.2 *Output Power.* The UMC shall be oscillated at each of the test frequencies in 4.6.8. The output power shall be maximized by:

(a) Adjusting the voltages supplied to the oscillator tube.
(b) Tuning the oscillator tube.
(c) Setting the SIG. SOURCE LEVEL SET for minimum attenuation.

The output power shall be measured at the SIG. SOURCE OUTPUT port using a dry calorimeter (drawing A10519341 or A105193-51, as required) and a calorimetric power meter (drawing A10528591) or approved equivalents and shall comply with 3.4.2.

4.6.3 *VSWR.* The VSWR shall be measured by the slotted section technique at each of the test frequencies in 4.6.8, and at the ports indicated in Table III, 3.4.3. When this test is performed at the SIG. SOURCE OUTPUT port, the RF LEVEL SET attenuator shall be set for minimum attenuation. When this test is performed at the FREQ CAL INPUT port, the FREQ METER LEVEL SET shall be set to approximately 40 db of attenuation. The maximum VSWR at each port shall comply with 3.4.3.

4.6.4 *Crystal Current.* The UMC's shall be oscillated at each of the test frequencies in 4.6.8. An output power level of 3 milliwatts minimum shall be established at the SIG. SOURCE OUTPUT port. Using a tuner at the FREQ CAL port, the crystal current indication on the crystal current meter of the UMC shall be maximized. With crystal current meter sensitivity set at maximum, the value of crystal current shall comply with 3.4.4.

4.6.5 *Rise and Fall Time of RF Notch.* A pulse of 1,000 cps repetition rate, 5 microsecond width and 0.015 microsecond rise and fall time shall be used to modulate the UMC's. With the CRYSTAL OUTPUT SELECTOR in EXT VIDEO position, the rise and fall time of the detected RF notch shall be displayed on an oscilloscope fed from the CRYSTAL CURRENT port of the UMC. The value of rise and fall time of the RF notch shall comply with 3.4.5.

4.6.6 *Detected Notch Amplitude on Oscilloscope.* A pulse of 1,000 cps repetition rate, 5 microseconds width and 0.015 microseconds rise and fall time shall be used to modulate the UMC's. With the CRYSTAL OUTPUT SELECTOR in EXT VIDEO position, the detected notch shall be displayed on an oscilloscope fed from the CRYSTAL CURRENT port of the UMC. The oscilloscope sensitivity shall be set at 0.01 volts/cm. The notch amplitude shall be maximized by:

(a) Adjusting the voltages supplied to the oscillator tube.
(b) Tuning the oscillator tube.
(c) Setting the SIG SOURCE LEVEL SET for minimum attenuation.

The unmodulated RF signal level of the SIGNAL SOURCE port shall be set to 1.0 milliwatt, for the MX–3175/USM–129 and MX–3175A/USM–129 and to 0.5 milliwatt for all the other UMC's by means of the dry calorimeter (drawings A10519341 or A105-19351, as required) and the calorimeter power meter (drawing A10528591) or approved equivalents. The detached notch amplitude shall comply with 3.4.6.

4.6.7 *Calibration Curve.* The insertion loss

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between the RF PULSE INPUT port and the SIGNAL SOURCE OUTPUT port shall be measured by the AF substitution technique using the attenuator calibrator (drawing A10519472) or approved equivalent. The calibration frequencies shall be as specified in table IV.

TABLE IV. *Calibration frequencies*

Designator	Calibration Points
MX–3297/USM–129	Every 50 mc
MX–3297A/USM–129	
MX–3169/USM–129	Every 100 mc
MX–3169A/USM–129	
MX–3170/USM–129	Every 200 mc
MX–3170A/USM–129	
MX–3171/USM–129	Every 400 mc
MX–3171A/UMS–129	
MX–3172/USM–129	Every 400 mc
MX–3172A/USM–129	
MX–3173/USM–129	Every 600 mc
MX–3173A/USM–129	
MX–3174/USM–129	Every 800 mc
MX–3174A/USM–129	
MX–3175/USM–129	Every 1000 mc
MX–3175A/USM–129	

A calibration curve of insertion loss versus frequency shall be prepared for each UMC.

4.6.8 *Test Frequencies.* The test frequencies for frequency range, output power, VSWR, crystal current, rise and fall time of RF notch and detected notch amplitude on oscilloscope shall be those specified in table V.

4.7 Environmental.

4.7.1 *Operating Tests.*

4.7.1.1 *Temperature.* When tested while exposed and thermally stabilized at ambient temperatures of plus 32° and plus 110° Fahrenheit, the UMC shall conform to the requirements of 3.4.

4.7.1.2 *Humidity.* After having been exposed to an atmosphere of 95 percent relative humidity at plus 77° Fahrenheit for 24 hours in nonoperating condition, the UMC, when tested while exposed to the same temperature and relative humidity, shall comply with the requirements of 3.4 for eight hours of continuous operation.

4.7.13 *Altitude.* When tested while exposed to a reduced pressure of 20.58 inches of mercury (equal to an altitude of 10,000 feet), the UMC shall comply with the requirements of 3.4 for four hours of continuous operation.

4.7.2 *Nonoperating Tests.*

4.7.2.1 *Temperature.* After having been stored for four hours while exposed and thermally stabilized at an ambient temperature of minus 25° Fahrenheit, the UMC shall be returned to normal conditions (plus 77° Fahrenheit, relative humidity 60 percent, sea level) and tested for compliance with 3.4. The test shall be repeated at an ambient temperature of plus 158° Fahrenheit.

4.7.2.2 *Altitude.* After having been stored for one hour while exposed to a reduced pressure of 3.44 inches of mercury (equal to an altitude of 50,000 feet), the UMC shall be returned to normal conditions and tested for compliance with 3.4.

4.7.2.3 *Vibration.* The UMC shall be properly secured in its combination case. It shall be subjected to a vibration whose frequency shall be varied linearly over the range from 5 to 52 cps with a double amplitude of 0.036 inches. The range shall be traversed from minimum to maximum frequency and returned to minimum frequency within one minute. The test shall be conducted for 15 minutes along each of three mutually perpendicular axes for a total of 45 minutes. At the conclusion of the test, the UMC shall show no physical damage and shall comply with 3.4.

4.7.2.4 *Shock.* The UMC shall be properly secured in its combination case. It shall be dropped from a height of one foot on the bottom face, a bottom edge, and a bottom corner. At the conclusion of the test, the UMC shall show no physical damage and shall comply with 3.4.

4.8 Sampling.

4.8.1 *Environmental Tests.* The environmental test of 4.7 shall be performed on a

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TABLE V. *Test frequencies (kmc/s)*

Parameter	MX-3297/ USM-129, MX-3297A/ USM-129	MX-3169/ USM-129, MX-3169A/ USM-129	MX-3170/ USM-129, MX-3170A/ USM-129	MX-3171/ USM-129, MX-3171A/ USM-129	MX-3172/ USM-129, MX-3172A/ USM-129	MX-3173/ USM-129, MX-3173A/ USM-129	MX-3174/ USM-129, MX-3174A/ USM-129	MX-3175/ USM-129, MX-3175A/ USM-129
Frequency Range	0.30 0.96	0.95 2.00	1.95 4.00	3.95 8.20	8.2 12.4	12.4 18.0	[2]18.0 22.0 [2]22.0 25.0 [3]24.0 26.5	[3]26.5 27.8 [4]27.8 32.2 [5]32.0 37.5 [6]35.0 40.0
Output Power,	0.30	0.9	1.95	3.95	8.2	12.4	18.00	26.50
VSWR, Crystal Current,	0.63	1.50	3.00	6.00	10.3	15.2	22.25	33.25
Rise and Fall Time of RF Notch, Detected Notch Ampl. on Osc.	0.96	2.00	4.00	8.20	12.4	18.0	26.50	40.00

[1] For MX-3176/USM-129 and MX-4254/USM-129A
[2] For MX-3177/USM-129 and MX-4255/USM-129A
[3] For MX-3178/USM-129 and MX-4256/USM-129A and MX-4488/USM-129A
[4] For MX-3179/USM-129 and MX-4257/USM-129A
[5] For MX-3180/USM-129 and MX-4258/USM-129A
[6] For MX-3181/USM-129 and MX-4259/USM-129A

representative UMC as specified by the contracting officer.

4.8.2 *Electrical Tests.* The electrical tests of 4.6 shall be performed on each UMC.

5. PREPARATION FOR DELIVERY

5.1 Packaging. No further packaging shall be required in addition to the combination case in which the UMC is housed. At least one of the two latches on each side of each combination case shall be secured with a security seal prior to shipment.

6. NOTES

6.1 Intended Use. The universal microwave calibrators and associated plug-in signal sources covered by this specification are part of Microwave Calibration Test Sets AN/USM–129, AN/USM–129A and AN/USM–129B. They are used in conjunction with other associated equipments of the Microwave Calibration Test Sets to perform the following measurements in the frequency range of 0.3 to 40.0 kmcps:

(a) The frequency measurement of an active source.
(b) The relative frequency response of a passive device.
(c) The peak power measurement of a pulse signal source.
(d) The measurement of average power, attenuation and voltage standing wave ratio.

The plus-in signal sources listed in table II, are required to cover the 18.0 to 26.5 kmcps and 26.5 to 40.0 kmcps frequency ranges respectively.

6.2 Ordering Data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Applicable stock number.

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(c) Provisions for initial production sample, including date of delivery (see 3.1).

(d) Environmental tests required (see 4.8.1).

6.3 Design Information. Design information regarding specific equipment not covered or referenced by this Specification may be obtained from the procuring activity upon specific request of the contractor.

Notice. When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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